

Derivation of Roe Flux for 1D Hyperbolic Conservation Laws

The Roe flux for a general 1D hyperbolic conservation law $\mathbf{u}_t + \mathbf{f}(\mathbf{u})_x = 0$ is:

$$\mathbf{F}_{i+1/2} = \frac{1}{2} (\mathbf{f}(\mathbf{u}_L) + \mathbf{f}(\mathbf{u}_R)) - \frac{1}{2} \sum_{k=1}^m |\tilde{\lambda}_k| \tilde{\alpha}_k \tilde{\mathbf{r}}_k$$

where $\mathbf{u}_L, \mathbf{u}_R$ are left and right states, $\tilde{\lambda}_k, \tilde{\alpha}_k, \tilde{\mathbf{r}}_k$ are Roe-averaged eigenvalues, wave strengths, and eigenvectors.

1. Burgers' Equation

Equation: $u_t + \left(\frac{u^2}{2}\right)_x = 0$.

1. Flux and Jacobian:

$$f(u) = \frac{u^2}{2}, \quad A = \frac{df}{du} = u$$

Eigenvalue: $\lambda = u$. Scalar system, eigenvector is 1.

2. Roe Averaging:

$$\tilde{u} = \frac{u_L + u_R}{2}, \quad \tilde{\lambda} = \tilde{u}$$

3. Wave Strength:

$$\Delta u = u_R - u_L, \quad \tilde{\alpha} = \Delta u$$

4. Roe Flux:

$$F_{i+1/2} = \frac{1}{2} \left(\frac{u_L^2}{2} + \frac{u_R^2}{2} \right) - \frac{1}{2} |\tilde{u}| (u_R - u_L)$$

2. Traffic Flow (Lighthill-Whitham-Richards) Equation

Equation: $\rho_t + (\rho v(\rho))_x = 0$, with $v(\rho) = v_{\max} \left(1 - \frac{\rho}{\rho_{\max}}\right)$.

1. Flux and Jacobian:

$$f(\rho) = \rho v_{\max} \left(1 - \frac{\rho}{\rho_{\max}}\right), \quad A = \frac{df}{d\rho} = v_{\max} \left(1 - \frac{2\rho}{\rho_{\max}}\right)$$

Eigenvalue: $\lambda = v_{\max} \left(1 - \frac{2\rho}{\rho_{\max}}\right)$. Eigenvector: 1.

2. Roe Averaging:

$$\tilde{\rho} = \frac{\rho_L + \rho_R}{2}, \quad \tilde{\lambda} = v_{\max} \left(1 - \frac{2\tilde{\rho}}{\rho_{\max}}\right)$$

3. Wave Strength:

$$\Delta \rho = \rho_R - \rho_L, \quad \tilde{\alpha} = \Delta \rho$$

4. Roe Flux:

$$F_{i+1/2} = \frac{1}{2} (f(\rho_L) + f(\rho_R)) - \frac{1}{2} \left| v_{\max} \left(1 - \frac{2\tilde{\rho}}{\rho_{\max}} \right) \right| (\rho_R - \rho_L)$$

3. Advection Equation

Equation: $u_t + (cu)_x = 0$, c constant.

1. Flux and Jacobian:

$$f(u) = cu, \quad A = \frac{df}{du} = c$$

Eigenvalue: $\lambda = c$. Eigenvector: **1**.

2. Roe Averaging:

$$\tilde{u} = \frac{u_L + u_R}{2}, \quad \tilde{\lambda} = c$$

3. Wave Strength:

$$\Delta u = u_R - u_L, \quad \tilde{\alpha} = \Delta u$$

4. Roe Flux:

$$F_{i+1/2} = \frac{1}{2} (cu_L + cu_R) - \frac{1}{2} |c| (u_R - u_L)$$

4. Nonlinear Scalar Conservation Law

Equation: $u_t + f(u)_x = 0$, $f(u)$ nonlinear (e.g., $f(u) = u^3$).

1. Flux and Jacobian:

$$A = \frac{df}{du} = f'(u)$$

Eigenvalue: $\lambda = f'(u)$. Eigenvector: **1**.

2. Roe Averaging:

$$\tilde{A} = \frac{f(u_R) - f(u_L)}{u_R - u_L} \quad (u_R \neq u_L), \quad \tilde{\lambda} = \tilde{A}$$

If $u_R = u_L$, use $\tilde{\lambda} = f'(u_L)$.

3. Wave Strength:

$$\Delta u = u_R - u_L, \quad \tilde{\alpha} = \Delta u$$

4. Roe Flux:

$$F_{i+1/2} = \frac{1}{2} (f(u_L) + f(u_R)) - \frac{1}{2} \left| \frac{f(u_R) - f(u_L)}{u_R - u_L} \right| (u_R - u_L)$$

5. p-System

Equation: $\mathbf{u} = [v, w]^T$, $\mathbf{f} = [w, p(v)]^T$, with $p(v) = -\sigma(v)$, $\sigma(v) = v^2$.

1. Flux and Jacobian:

$$\mathbf{f} = \begin{bmatrix} w \\ -v^2 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2v & 0 \end{bmatrix}$$

Eigenvalues: $\lambda_1 = -\sqrt{2v}$, $\lambda_2 = \sqrt{2v}$. Eigenvectors:

$$\mathbf{r}_1 = \begin{bmatrix} 1 \\ -\sqrt{2v} \end{bmatrix}, \quad \mathbf{r}_2 = \begin{bmatrix} 1 \\ \sqrt{2v} \end{bmatrix}$$

2. Roe Averaging:

$$\tilde{v} = \sqrt{v_L v_R}, \quad \tilde{w} = \frac{\sqrt{v_L} w_L + \sqrt{v_R} w_R}{\sqrt{v_L} + \sqrt{v_R}}$$

Eigenvalues: $\tilde{\lambda}_1 = -\sqrt{2\tilde{v}}$, $\tilde{\lambda}_2 = \sqrt{2\tilde{v}}$. Eigenvectors:

$$\tilde{\mathbf{r}}_1 = \begin{bmatrix} 1 \\ -\sqrt{2\tilde{v}} \end{bmatrix}, \quad \tilde{\mathbf{r}}_2 = \begin{bmatrix} 1 \\ \sqrt{2\tilde{v}} \end{bmatrix}$$

3. Wave Strengths:

$$\Delta \mathbf{u} = \begin{bmatrix} v_R - v_L \\ w_R - w_L \end{bmatrix} = \tilde{\alpha}_1 \tilde{\mathbf{r}}_1 + \tilde{\alpha}_2 \tilde{\mathbf{r}}_2$$

$$\tilde{\alpha}_1 = \frac{\Delta w + \sqrt{2\tilde{v}} \Delta v}{2\sqrt{2\tilde{v}}}, \quad \tilde{\alpha}_2 = \frac{\Delta w - \sqrt{2\tilde{v}} \Delta v}{-2\sqrt{2\tilde{v}}}$$

where $\Delta v = v_R - v_L$, $\Delta w = w_R - w_L$.

4. Roe Flux:

$$\mathbf{F}_{i+1/2} = \frac{1}{2} \left(\begin{bmatrix} w_L \\ -v_L^2 \end{bmatrix} + \begin{bmatrix} w_R \\ -v_R^2 \end{bmatrix} \right) - \frac{1}{2} \left(|\tilde{\lambda}_1| \tilde{\alpha}_1 \tilde{\mathbf{r}}_1 + |\tilde{\lambda}_2| \tilde{\alpha}_2 \tilde{\mathbf{r}}_2 \right)$$

6. Keyfitz-Kranzer System

Equation: $\mathbf{u} = [u, v]^T$, $\mathbf{f} = [u\sqrt{u^2 + v^2}, v\sqrt{u^2 + v^2}]^T$.

1. Flux and Jacobian: Let $r = \sqrt{u^2 + v^2}$.

$$\mathbf{f} = \begin{bmatrix} ur \\ vr \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} \frac{2u^2+v^2}{r} & \frac{uv}{r} \\ \frac{uv}{r} & \frac{u^2+2v^2}{r} \end{bmatrix}$$

Eigenvalues: $\lambda_1 = 0$, $\lambda_2 = 2r$. Eigenvectors:

$$\mathbf{r}_1 = \begin{bmatrix} -v \\ u \end{bmatrix}, \quad \mathbf{r}_2 = \begin{bmatrix} u \\ v \end{bmatrix}$$

2. Roe Averaging:

$$\tilde{u} = \frac{\sqrt{r_L}u_L + \sqrt{r_R}u_R}{\sqrt{r_L} + \sqrt{r_R}}, \quad \tilde{v} = \frac{\sqrt{r_L}v_L + \sqrt{r_R}v_R}{\sqrt{r_L} + \sqrt{r_R}}$$

with $r_L = \sqrt{u_L^2 + v_L^2}$, $r_R = \sqrt{u_R^2 + v_R^2}$, $\tilde{r} = \sqrt{\tilde{u}^2 + \tilde{v}^2}$. Eigenvalues: $\tilde{\lambda}_1 = 0$, $\tilde{\lambda}_2 = 2\tilde{r}$. Eigenvectors:

$$\tilde{\mathbf{r}}_1 = \begin{bmatrix} -\tilde{v} \\ \tilde{u} \end{bmatrix}, \quad \tilde{\mathbf{r}}_2 = \begin{bmatrix} \tilde{u} \\ \tilde{v} \end{bmatrix}$$

3. Wave Strengths:

$$\Delta \mathbf{u} = \begin{bmatrix} u_R - u_L \\ v_R - v_L \end{bmatrix} = \tilde{\alpha}_1 \tilde{\mathbf{r}}_1 + \tilde{\alpha}_2 \tilde{\mathbf{r}}_2$$

$$\tilde{\alpha}_1 = \frac{-\tilde{u}\Delta v + \tilde{v}\Delta u}{\tilde{u}^2 + \tilde{v}^2}, \quad \tilde{\alpha}_2 = \frac{\tilde{u}\Delta u + \tilde{v}\Delta v}{\tilde{u}^2 + \tilde{v}^2}$$

4. Roe Flux:

$$\mathbf{F}_{i+1/2} = \frac{1}{2} \left(\begin{bmatrix} u_L r_L \\ v_L r_L \end{bmatrix} + \begin{bmatrix} u_R r_R \\ v_R r_R \end{bmatrix} \right) - \frac{1}{2} (|0| \tilde{\alpha}_1 \tilde{\mathbf{r}}_1 + |2\tilde{r}| \tilde{\alpha}_2 \tilde{\mathbf{r}}_2)$$

Since $|0| = 0$:

$$\mathbf{F}_{i+1/2} = \frac{1}{2} (\mathbf{f}(\mathbf{u}_L) + \mathbf{f}(\mathbf{u}_R)) - \tilde{r} \tilde{\alpha}_2 \begin{bmatrix} \tilde{u} \\ \tilde{v} \end{bmatrix}$$

Each derivation ensures a conservative, consistent flux satisfying the Rankine-Hugoniot condition.